SHRUTI VATSA

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RESEARCH INTERESTS

Foraging and uncertainty · Cognitive mechanisms of information search · Decision Neuroscience · Computational modelling · Bayesian Inference

EDUCATION

BTech - MTech (Dual Degree) **Indian Institute of Technology Kanpur**
B.Tech Major: Civil Engineering(CE) *Jul 2018 - Dec 2023*
M.Tech Major: Biological Sciences and Bioengineering(BSBE)

Thesis: *Influence of environmental structure on foraging behaviour* (Supervisor: Dr. Arjun Ramakrishnan)

RESEARCH EXPERIENCE

Junior Research Fellow
Supervised by Dr. Saurabh Gandhi, Cognition & Neural Dynamics Lab, IIT Jodhpur *Feb 2025 - Jul 2025*

- Conducted an extensive literature review on digital foraging, affective states, and decision-making.
- Designed the preliminary version of the task interface and experimental flow for studying smartphone-based information foraging.

Project Engineer
Supervised by Dr. Arjun Ramakrishnan, Decision Lab, IIT Kanpur *Jan 2024 - Dec 2024*

- Investigated the effect of structural uncertainty of the environment on decision-making as a function of anxiety and depression scores assessed through standard mental health questionnaires using a patch-foraging task.
- Modelled data with generalized linear mixed-effects models (GLMMs), linear mixed-effects models (LMMs), hierarchical drift diffusion models (HDDMs), principal component analysis (PCA), and bayesian regression models (BRMS).

Master's thesis: Influence of environmental structure on foraging behaviour
Supervised by Dr. Arjun Ramakrishnan, IIT Kanpur *May 2022 - Dec 2023*

- Co-designed a 2D web-based foraging task to understand how environment structure, anxiety, and depression influence patch-leaving behaviour.
- Applied LMM and HDDM frameworks to model optimality deviations.

Research Internship
WeUnlearn *Mar 2021 - Apr 2021*

- Conducted behavioral studies on mental health interventions via chatbots for adolescents.
- Project placed 3rd in the Stanford Center on Longevity Design Challenge.

Summer Internship **[presentation]**
Project mentored by Dr. Satish B. Agnihotri, IIT Bombay *May 2019 - July 2019*

- Worked on creation of a spatial data-driven decision support mechanism for women and child development department of Palghar district for Zilla Parishad, Palghar district, Maharashtra using QGIS and TABLEAU.

PUBLICATIONS & TALKS

Vatsa S., Biswal A., Ramakrishnan A. (*in preparation*). Effect of structure and trait anxiety on reward-guided decision-making in a patch-foraging environment
Vatsa S., Ramakrishnan A. (*oral presentation*). Influence of structure and trait-anxiety on patch foraging behaviour. Oral presentations at:

- *Annual Conference of Cognitive Science 11th Edition, IIT Bombay, December 2024*
- *2024 Indian Conference on Judgement and Decision Making, IIT Mandi, October 2024*

WORKSHOPS & TECHNICAL EVENTS

5TH Brain Computation and Learning workshop (attended)
Hosted at Indian Institute of Science, Bangalore — *Jun 30 - Jul 04, 2025*

9TH Annual Conference of Cognitive Science (attended)
Hosted at Indian Institute of Technology Delhi — *Dec 8 - Dec 10, 2022*

Hands-on HCI and BCI Workshop (attended)
Workshop, Hosted by Upside Down Labs at SIIC IIT Kanpur — *October 2, 2022*

Computational Neuroscience Workshop (attended, virtual) — **[certificate]**
Interactive Track, Neuromatch Academy — *July 2023*

Census GIS Workshop (attended)
Hosted at Indian Institute of Technology Bombay — *June 24 - 26, 2019*

TEACHING & MENTORSHIP

Teaching Assistant
Computational Neuroscience, Neuromatch Academy — *July 2024*
Guided 17 students in the computational neuroscience training program which utilised hands-on Python programming through tutorials and research projects.

Teaching Assistant
Decision-making and The Brain(BSE662A), IIT Kanpur — *Jan 2024 - April 2024*
Aided in smooth functioning of the class, course exams, and group research projects.

Academic Mentor
Counselling Service, IIT Kanpur — *July 2019 - April 2020*
Conducted institute-level remedial classes and provided one-on-one academic assistance to the freshers for the course *Engineering Mechanics(PHY102A)*.

LEADERSHIP & OUTREACH

Co-founder
Chuckles Art Studio — *Jan 2024 - Present*
Art and design startup creating literature and science inspired t-shirts, hoodies, and wall art.

Founding Member
IIT Kanpur Women's Football Team — *2019 - 2022*
Initiated team formation; won 2 Bronze & 1 Silver medals at Udghosh, IITK's national sports fest.

Student Guide
Counselling Service, IIT Kanpur — *2019 - 2020*

Helped organize the weeklong annual orientation for 1000+ freshmen.

Media and Publicity Lead
Entrepreneurship Cell, IIT Kanpur *2019 - 2021*
Led a 10 member team to manage the media coverage of entrepreneurial events.

RELEVANT COURSEWORK

Statistics/Programming	Fundamentals of Computing, Computational Methods in Engineering, Probability & Statistics, Statistics for Modern Biology
Cognitive Science	Computational Cognitive Science, Cognitive Neuroscience, Decision Making and the Brain
Biology	Bioinformatics and Computational Biology, Biochemistry, Structural Biology, Modern Instrumental Methods in Biological Sciences

TECHNICAL SKILLS

Programming	Python & R (*proficient*). MATLAB, HTML, JavaScript, C/C++ (*familiar*).
Modeling and Analysis	LMM, GLMM (*lme4, glmmTMB*), HDDM, Bayesian inference (*brms*), PCA.
Software/utilites	LaTeX, Qualtrics, Amazon Web Services(S3 and Lambda), Psychopy, QGIS, Tableau, Balsamiq Wireframes, Figma, Autodesk Inventor, Ahrefs.
Bioinformatics skills	Blast, Clustal, PDB, RasMol.